

Worksheet

Evaluate the following integrals.

1. $\int \frac{x^2 + x - 5}{x + 3} dx =$

2. $\int \cos x dx =$

3. $\int e^{3x} \sin(2x) dx =$

4. $\int \frac{1}{\sqrt{1-x^2}} dx =$

5. $\int \frac{1}{a^2 + x^2} dx =$

6. $\int x e^{3x} dx =$

7. $\int \sqrt{x} \ln(5x) dx =$

8. $\int \frac{x + 5}{x^2 + 6x + 10} dx =$

9. $\int \cot x dx =$

10. $\int \frac{1}{\sqrt{x^2 - 7}} dx$

11. $\int \frac{x - 1}{(x + 1)(x^2 + 3x + 4)} dx =$

12. $\int e^x dx =$

13. $\int x^3 \cos(2x) \, dx =$

14. $\int x^e \, dx =$

15. $\int \cos^4(3x) \, dx$

16. $\int \frac{3x+2}{(x-1)(x+4)} \, dx =$

17. $\int \frac{1}{\sqrt{x^2+6x+34}} \, dx$

18. $\int \sec x \, dx =$

19. Set up (do not solve) the partial fraction decomposition for $\frac{4x-3}{x^2(x+3)(x^2+2x+9)(x^2+1)^3}$.

20. Does the sequence $\{\sqrt[n]{n}\}$ converge? If so, what is the limit? If not, why not?

21. Find $\lim_{n \rightarrow \infty} \left(1 - \frac{3}{n}\right)^{2n}$.

22. Find $\lim_{n \rightarrow \infty} 3^{1/n}$.

23. Find $\lim_{n \rightarrow \infty} r^n$, where r is some real number. Does your answer depend on what r is? If so, explain how. If not, explain why.

24. Suppose $\{a_n\}$ is a sequence of positive numbers and $a_n < \frac{1}{n^2}$. Must $\{a_n\}$ converge? Why or why not?